Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		• Can be used in {all organisms/cells/species}/ Can be used in both prokaryotes and eukaryotes (1) • For {most/ all} {reactions/ biological processes/ metabolic processes} (1)	2			2		
		(ii)		Glucose and water / accept correct chemical formulae (1) ADP and {P_i / phosphate} (1) <u>Contracted/ contracting}</u> muscle (1)		3		3		
	(b)	(i)		A-chloroplast, B-mitochondria (1) **Both correct for 1 mark**		1		1		
		(ii)		**Organelle A** / **Organelle B**: thylakoid / Inner membrane (1) Thylakoid space/lumen / Matrix (1) stroma / Intermembrane space (1) **1 mark for each correct row** Ecf from (i)		3		3		
		(iii)		**Any five (×1) from:** A. Proton pump pumps protons {into intermembrane space/ across membrane/ from Y to Z} (1) B. To create {proton / H^+/electrochemical} gradient (1) C. {ATP synth(et)ase/ stalked particle} allows {protons / H^+s} {through membrane/ down gradient} (1) D. {convert/combine} ADP (+ P_i) to ATP (1) E. correct reference to {electron carriers/ electron transport chain} (1) F. Ref to: Proton pump **and** ATP synth(et)ase (1)	3	2		5		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			A. In pH 4 (no ATP produced as) no proton gradient (1) B. When pH {raised/8} a proton gradient is produced / The buffer solution at pH 8 has a lower proton concentration than the vesicle / ORA (1) C. (When the proton gradient is in place) protons move {out/ across the membrane/ through stalked particle/ through ATP synth(et)ase} (1) D. (When protons move across the membrane) ATP is made / no ATP is made when protons do not move across (1)			4	4		4
				Question 3 total	**5**	**9**	**4**	**18**	**0**	**4**